WATER RESOURCES, WELLS & PUMPS · NOTE WR02

Sources of Irrigation Water and the River Systems of India

Course topics: IDE5.5 L2, IDE8.16 L4

Sources of water

Surface sources	Sub-surface (groundwater) sources
Natural ponds and lakes — natural depressions filled with water	**Springs** — natural outflow where a pervious layer lies between two impervious layers; small supplies
Streams and rivers — the most important public source; carry silt and suspended matter, so need treatment	**Infiltration wells** — shallow wells along river banks collecting water seeping through the bed
Impounding reservoirs — artificial lakes upstream of dams (dam or river reservoirs); supply water all year	**Infiltration galleries** (written "infiltration glaciers" in the notes) — horizontal or nearly horizontal tunnels at 3–5 m depth along river banks through water-bearing strata; sometimes called horizontal wells
Oceans — hold most water but are generally not used for supply	**Wells** — open wells (large diameter, low discharge of 3–6 L/s) and tubewells (deep drilled pipes, higher discharge, tapping several water-bearing strata)

Groundwater lies in the saturated zone, in the pores and cracks of rocks; how much occurs depends on porosity and permeability.

How water reaches the fields

1. Store river flood water behind **dams** at suitable sites, or divert it with **weirs or barrages**.
2. Collect runoff from small areas in **ponds or tanks**.
3. Tap underground reservoirs with **wells** fitted with pumps or water lifts.
4. Convey water from reservoirs or diversion points through **canals**, which run from higher to lower ground under gravity.

Class by source level	How water reaches the field
Flow irrigation	By gravity from a source above the field
Lift irrigation	Raised or lifted from a source below the field

Important structures

Structure	Facts (as in the source notes)
Bhakra Dam	One of the world's highest straight gravity concrete dams; completed 1962; height 225.55 m; width 9.14 m at the top and 190.50 m at the base. Reservoir: Govind Sagar, 96.56 km long, water spread 168.5 sq km, storage capacity 9,87,000 ha-m
Weir on the Mahi	At Wanakbori near Vadodara, Gujarat; diverts flow into the Mahi Right Bank Canal system; river flow is augmented by releases from Kadana reservoir 65 km upstream
Main canal	Brick-lined main canal of a major irrigation project

Other published figures for Bhakra: completed in 1963, height 226 m, reservoir area 168.35 km², gross storage about 9.34 billion m³ (≈ 934,000 ha-m). Figures differ by source and reference level — check the Bhakra Beas Management Board before quoting.

River systems of India

Group	Character
Himalayan rivers	Fed by rain and by melting snow and glaciers; flow nearly all year; swell and flood in the monsoon
Deccan rivers	Mostly rain-fed; large seasonal fluctuation; many non-perennial
Coastal rivers	Especially on the west coast; short, small catchments; mostly non-perennial
Inland drainage rivers	Mainly western Rajasthan; drain into salt lakes (e.g. Sambhar) or disappear into the sand with no outlet to the sea

- Most rivers flow into the Bay of Bengal; some flow west into the Arabian Sea. Inland drainage occurs in the northern Aravalli range, parts of Ladakh and the Thar Desert.
- Indian rivers rise in three main watersheds: the Himalaya and Karakoram; the Chota Nagpur plateau and Vindhya–Satpura ranges; and the Western Ghats.

River	Key fact
Ganga	Longest river (2525 km); rises at the Gangotri Glacier
Yamuna	Rises at the Yamunotri Glacier
Brahmaputra	Rises in Tibet; enters India in Arunachal Pradesh
Indus	Rises on the Tibetan Plateau; drains into the Arabian Sea
Godavari	Longest peninsular river
Krishna, Kaveri, Narmada, Tapi	Major river systems
Sabarmati	Rises near Udaipur, Rajasthan
Luni	West-flowing river of inland drainage (Rajasthan)

River basins

A river basin is the whole area drained by a river and its tributaries.

Class	Catchment area
Large	≥ 20,000 km²
Medium	2,000 to 20,000 km²
Minor	< 2,000 km²

- India has 20 major river basins and 45 medium basins — 24 east-flowing, 17 west-flowing and 4 draining into neighbouring countries (north-east).
- Major and medium basins yield over 90% of rainfall runoff; minor rivers only about 10%.
- Central Water Commission (2013): total average water resources 1869.37 BCM; utilisable surface water 690.1 BCM. The biggest contributors are the Ganga–Brahmaputra–Meghna system, Godavari, Krishna, Mahanadi and the west-flowing rivers. (CWC's 2019 reassessment: 1999.20 BCM.)

Sources: Central Water Commission, Reassessment of Water Availability in India using Space Inputs (2019); Rajya Sabha reply (Dec 2022) on per-capita availability; CGWB Dynamic Ground Water Resources of India (2023 and 2025 assessments); CGWB & Ground Water Department, Govt of Rajasthan, Dynamic Ground Water Resources of Rajasthan 2024; WHO fact sheet 'Diarrhoeal disease' (7 March 2024). ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Irrigation in which water reaches the field by gravity from a higher source is

(a) lift irrigation (b) flow irrigation
(c) sub-irrigation (d) drip irrigation

Q2. Rivers that drain into salt lakes or vanish into sand are found mainly in

(a) Kerala (b) western Rajasthan
(c) Assam (d) Odisha

Q3. A river basin with catchment ≥ 20,000 km² is classed as

(a) minor (b) medium
(c) large (d) composite

Q4. The longest peninsular river of India is the

(a) Krishna (b) Narmada
(c) Godavari (d) Kaveri

Q5. The reservoir of the Bhakra dam is

(a) Gandhi Sagar (b) Govind Sagar
(c) Rana Pratap Sagar (d) Indira Sagar

Q6. Open wells typically give a low discharge of about

(a) 0.3–0.6 L/s (b) 3–6 L/s
(c) 30–60 L/s (d) 300 L/s

Answers

Q1 — (b) flow irrigation
Q2 — (b) western Rajasthan
Q3 — (c) large
Q4 — (c) Godavari
Q5 — (b) Govind Sagar

Q6 — (b) 3–6 L/s